

ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Term Project Preliminary Results

Purpose

The preliminary results assignment should demonstrate that you can run a complete modeling workflow and interpret the initial outputs. The analysis does not need to be final. You may revise your data, model settings, evaluation approach, or figures after receiving feedback.

You will be evaluated on whether the analysis is appropriate, reproducible, and interpreted carefully. You will not be evaluated on whether the model produces high accuracy or supports your expected conclusion.

Required content

1. Research question and analysis status

Restate your research question and briefly describe any changes made since the data and modeling plans. Identify which parts of the analysis are complete and which remain preliminary.

2. Data used

Summarize the data included in the preliminary analysis:

- Number of occurrence records before and after filtering
- Geographic and temporal coverage
- Definition of M
- Number and sampling method for background locations
- Environmental predictors included
- Any predictors removed and the reason for removing them

Report important limitations or unresolved data problems.

3. Preliminary model

Describe the model or models you fitted. Include:

- Modeling algorithm
- Predictor set
- Settings controlling model complexity
- Data-partitioning method
- Any tuning or parameter selection
- Calibration and projection regions

Provide enough information for someone familiar with the method to understand the analysis.

4. Model evaluation

Report the evaluation measures used and explain what they assess. Include results from withheld or partitioned data when possible. Discuss whether spatial clustering, sampling bias, small sample size, or environmental novelty could influence the evaluation. Do not interpret strong evaluation scores as evidence that the model represents the species' complete ecological niche.

5. Preliminary figures and results

Include at least four clearly labeled outputs:

- A map showing occurrences and the proposed M(s)
- A preliminary relative-suitability or environmental-association output
- A model-evaluation
- A robustness comparison

You may also include response curves, variable importance, environmental-space plots, or maps comparing model assumptions. Each figure should have a caption explaining what it shows and why it matters for the research question.

6. Biological interpretation

State one preliminary biological interpretation supported by the results. Explain:

- What pattern provides the evidence
- Whether the interpretation concerns environmental association, relative suitability, or another model output
- What the model cannot establish
- Whether the result agrees with your biological expectations

Unexpected or weak results are acceptable when they are explained carefully.

7. Robustness and uncertainty

Present results from at least one alternative modeling decision. This could involve changing:

- M
- Predictor set
- Background sample
- Occurrence filtering
- Model algorithm or complexity
- Data partitioning

Describe the interpretation as stable, conditional, or not robust across the alternatives tested. Identify additional assumptions that still need evaluation.

8. Next steps

Identify the most important changes you plan to make before the final analysis. Focus on specific actions, such as revising M, removing a predictor, changing model complexity, improving evaluation, or checking environmental novelty.

Reproducibility requirements

Your Term Project repository should contain:

- Scripts that run from the project root
- Clearly organized data and output folders
- Model settings and random seeds
- Figures included in the report
- An updated README explaining how to reproduce the preliminary analysis
- A current GitHub version with all required files pushed

Format and submission

- Due Friday, November 6 by 11:59 PM
- Up to five pages, including figures
- PDF format
- 12-point Times New Roman
- Include citations for data and modeling methods
- Submit the PDF and private GitHub repository link through Canvas